

Relazione-4

Setup del cluster.

Dopo il scaricamento di FortiClient e la configurazione seguita sulla consegna sono riuscito a connettermi con il comando SSH al cluster di ateneo. Ecco un screenshot:

```
→ 4-Setup ssh kennethjames.nnaminkousse@login.hpc.unipr.it
kennethjames.nnaminkousse@login.hpc.unipr.it's password:
Last login: Thu Mar 12 11:39:05 2026 from ui03
```

Università degli Studi di Parma - Infrastruttura HPC

login.hpc.unipr.it

<https://www.hpc.unipr.it>

/hpc/archive:		64.0%
/hpc/group :		58.6%
/hpc/home :		97.6%
/hpc/scratch:		81.8%
/hpc/share :		91.2%

/hpc/scratch: il 17/03/2026 i file non acceduti da più di 90 giorni verranno rimossi (cancellati). Gli utenti sono comunque invitati a rimuovere autonomamente i file non più utilizzati.

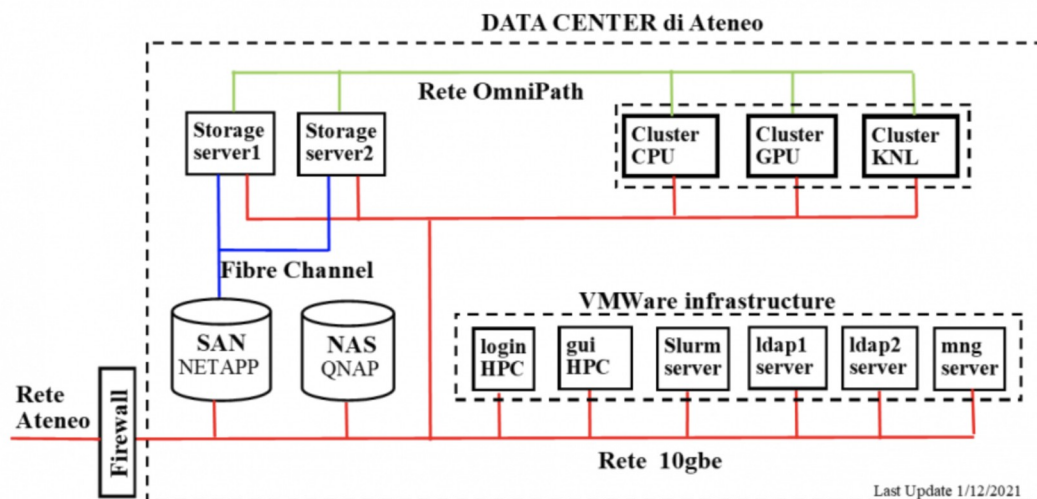
/hpc/scratch: on 17/03/2026 files not accessed for more than 90 days will be removed (deleted). Users are however advised to independently remove files that are no longer used.

```
[kennethjames.nnaminkousse@ui01 ~]$ |
```

2026-03-12 20:31

Adesso ecco lo screenshot del cluster di ateneo:

hpcschema2021.png



Primo Job Su CPU

Qui eseguendo il file lsCPU.slurm ecco il risultato:

→ sbatch lsCPU.slurm

```
SLURM_JOBID=2433415
SLURM_NODELIST=wn99
SLURM_JOB_NAME=lsCPU.slurm
SLURM_MEM_PER_CPU=512
SLURM_NNODES=1
SLURM_TASKS_PER_NODE=1
SLURM_JOB_USER=kennethjames.nnaminkousse
SLURM_SUBMIT_DIR=/hpc/home/kennethjames.nnaminkousse/HPC2526/deviceQuery/CPU
SLURM_CPUS_ON_NODE=1
SLURM_JOB_CPUS_PER_NODE=1
SLURM_JOB_CPUS_PER_NODE=cpu_guest
SLURM_JOB_ACCOUNT=t_2025_hpcprogrpar
SLURM_JOB_NUM_NODES=1
wn99
Architecture:           x86_64
CPU op-mode(s):         32-bit, 64-bit
Byte Order:              Little Endian
CPU(s):                  64
On-line CPU(s) list:     0-31
Off-line CPU(s) list:    32-63
Thread(s) per core:      1
Core(s) per socket:      16
Socket(s):                2
NUMA node(s):            2
Vendor ID:               AuthenticAMD
CPU family:               23
Model:                   49
Model name:               AMD EPYC 7282 16-Core Processor
Stepping:                 0
CPU MHz:                  1500.000
CPU max MHz:              2800.0000
CPU min MHz:              1500.0000
BogoMIPS:                 5589.67
Virtualization:           AMD-V
L1d cache:                32K
L1i cache:                32K
L2 cache:                 512K
L3 cache:                 16384K
NUMA node0 CPU(s):        0-15
NUMA node1 CPU(s):        16-31
Flags:                    fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush
tsc extd_apicid aperfmperf eagerfpu pni pclmulqdq monitor ssse3 fma cx16 sse4_1 sse4_2 x2apic movbe
ch osvw ibs skinit wdt tce topoext perfctr_core perfctr_nb bpext perfctr_l2 cpb cat_l3 cdp_l3 hw_p
flushopt clwb sha_ni xsaveopt xsavec xgetbv1 cqm_llc cqm_occup_llc cqm_mbm_total cqm_mbm_local cl
ter pfthreshold avic v_vmsave_vmload vgif umip overflow_recov succor smca
```

Si vede che il JOBID e 2433415 e il nodo su qui e stato eseguito lo script e wn99.

→ Ha 2 sockets, e 16 cores per sockets quindi 32 cores in totale.

Tutto questio e letto dal file output lsCPU.slurm. 2433415.out

```
[kennethjames.nnaminkousse@ui03 CPU]$ ls -lt
total 2
-rw-r--r-- 1 kennethjames.nnaminkousse Person 2079 Mar 29 12:43 lsCPU.slurm.2433415.out
-rwxr-xr-x 1 kennethjames.nnaminkousse Person 1604 Feb 19 18:35 lsCPU.slurm
-rwxr-xr-x 1 kennethjames.nnaminkousse Person 155 Feb 19 18:34 lsCPU.sh
```

Primo Job GPU

Ho eseguito lo script deviceQuery.slurm del GPU usando un GPU di tipo P100 e ecco il risultato :

```
module
wn41
GPU 0: Tesla P100-PCIE-12GB (UUID: GPU-712c408a-aea1-bedc-0017-e8b596a19813)
./deviceQuery Starting...

    CUDA Device Query (Runtime API) version (CUDA static linking)

Detected 1 CUDA Capable device(s)

Device 0: "Tesla P100-PCIE-12GB"
  CUDA Driver Version / Runtime Version      12.4 / 11.5
  CUDA Capability Major/Minor version number: 6.0
  Total amount of global memory:             12187 MBytes (12778733568 bytes)
  (056) Multiprocessors, (064) CUDA Cores/MP: 3584 CUDA Cores
  GPU Max Clock rate:                       1329 MHz (1.33 GHz)
  Memory Clock rate:                        715 Mhz
  Memory Bus Width:                         3072-bit
  L2 Cache Size:                           3145728 bytes
  Maximum Texture Dimension Size (x,y,z)     1D=(131072), 2D=(131072, 65536), 3D=(16384, 16384, 16384)
  Maximum Layered 1D Texture Size, (num) layers 1D=(32768), 2048 layers
  Maximum Layered 2D Texture Size, (num) layers 2D=(32768, 32768), 2048 layers
  Total amount of constant memory:           65536 bytes
  Total amount of shared memory per block:    49152 bytes
  Total shared memory per multiprocessor:     65536 bytes
  Total number of registers available per block: 65536
  Warp size:                                32
  Maximum number of threads per multiprocessor: 2048
  Maximum number of threads per block:        1024
  Max dimension size of a thread block (x,y,z): (1024, 1024, 64)
  Max dimension size of a grid size   (x,y,z): (2147483647, 65535, 65535)
  Maximum memory pitch:                     2147483647 bytes
  Texture alignment:                         512 bytes
  Concurrent copy and kernel execution:      Yes with 2 copy engine(s)
  Run time limit on kernels:                 No
  Integrated GPU sharing Host Memory:         No
  Support host page-locked memory mapping:   Yes
  Alignment requirement for Surfaces:        Yes
  Device has ECC support:                    Enabled
  Device supports Unified Addressing (UVA):   Yes
  Device supports Managed Memory:            Yes
  Device supports Compute Preemption:        Yes
  Supports Cooperative Kernel Launch:        Yes
  Supports MultiDevice Co-op Kernel Launch:  Yes
  Device PCI Domain ID / Bus ID / location ID: 0 / 5 / 0
  Compute Mode:
    < Default (multiple host threads can use ::cudaSetDevice()) with device simultaneously >

deviceQuery, CUDA Driver = CUDART, CUDA Driver Version = 12.4, CUDA Runtime Version = 11.5, NumDevs = 1
Result = PASS
```

- Il nodo di calcolo è wn41 e ha 3584 CUDA CORES
- ha un Max Clock rate di circa 1.33GHz.

Laboratorio 2 (MatrixMul e Network con I PLOTTINGS)

MatrixMul:<Imagine png>

Network:<Imagine png>